Linux File Permissions

Day 1 Investigation Worksheet — detailed responses

1. What does rwx mean?

Item	Response
What did I ask?	What does `rwx` mean in Linux?
What did AI tell me?	`rwx` means read (r), write (w), and execute (x) permissions in Linux/Unix file systems.
What did I expect?	Linux filesystem permissions.
What command/action did I perform?	`ls -l`
What happened?	Printed a long list with items with their permission details.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Relevance of read, write and execute permissions in Linux.

2. What are owner, group and others?

Item	Response
What did I ask?	What are owner, group and others?
What did AI tell me?	Owner – User who owns the file; Group – Users who belong to the file's assigned group; Others – Everyone else who is not the owner or part of that group.
What did I expect?	Users tied to Linux filesystem items.
What command/action did I perform?	`ls -l` <file_name>
What happened?	Printed the file's user permissions – `.rw-r--r--`
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Identifying owner, group and others permissions for a Linux item.

3. What does chmod 755 mean?

Item	Response
What did I ask?	What does `chmod 755` mean?
What did AI tell me?	`chmod 755` gives the owner read, write, execute (`rwx`), and group and others read and execute (`r-x`) permissions.
What did I expect?	r = 4; w = 2; x = 1; Owner will get 4 + 2 + 1 = 7 {`rwx`}; Group and Others will get 4 + 1 = 5 {`r-x`}.
What command/action did I perform?	`chmod 755` <file_name> ; `ls -l` <file_name>
What happened?	Assigned 755 permission to <file_name>; printed the file's permissions `.rwxr-xr-x`
Yes / No / Partially (AI correctness)	Yes
What did I learn?	To set user permissions to Linux filesystem items.

4. What does chmod 644 mean?

Item	Response
What did I ask?	What does `chmod 644` mean?
What did AI tell me?	`chmod 644` gives the owner read and write (`rw-`) permissions, and group and others read-only (`r--`) permissions.
What did I expect?	r = 4; w = 2; x = 1; Owner will get 4 + 2 = 6 {`rw-`}; Group and Others will get 4 {`r--`}.
What command/action did I perform?	`chmod 644` <file_name> ; `ls -l` <file_name>
What happened?	Assigned 644 permission to <file_name>; printed the file's permissions `.rw-r--r--`
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Identifying owner, group and others permissions for a Linux item.

5. Why is chmod 777 generally discouraged?

Item	Response
What did I ask?	Why is `chmod 777` generally discouraged?
What did AI tell me?	`chmod 777` is discouraged because it gives everyone read, write, and execute permissions, allowing any user to modify, delete, or execute the file, creating serious security risks.
What did I expect?	Giving {`rwx`} permission to owner, group and others will cause Linux item boundary breach.
What command/action did I perform?	`chmod 777` <file_name> ; `rm -r` <file_name>
What happened?	Assigned 777 permission to <file_name>; no user operation restrictions were applied to the <file_name>.
Yes / No / Partially (AI correctness)	Yes
What did I learn?	Identifying owner, group and others permissions for a Linux item.

Challenge question

Q: Is `chmod 777` **always required when a user gets "Permission denied"?**

A: No. Permission denied can often be fixed by using the correct owner, group, or permissions (e.g., `chmod 755` or `chown`); avoid `chmod 777` because it creates security risks.